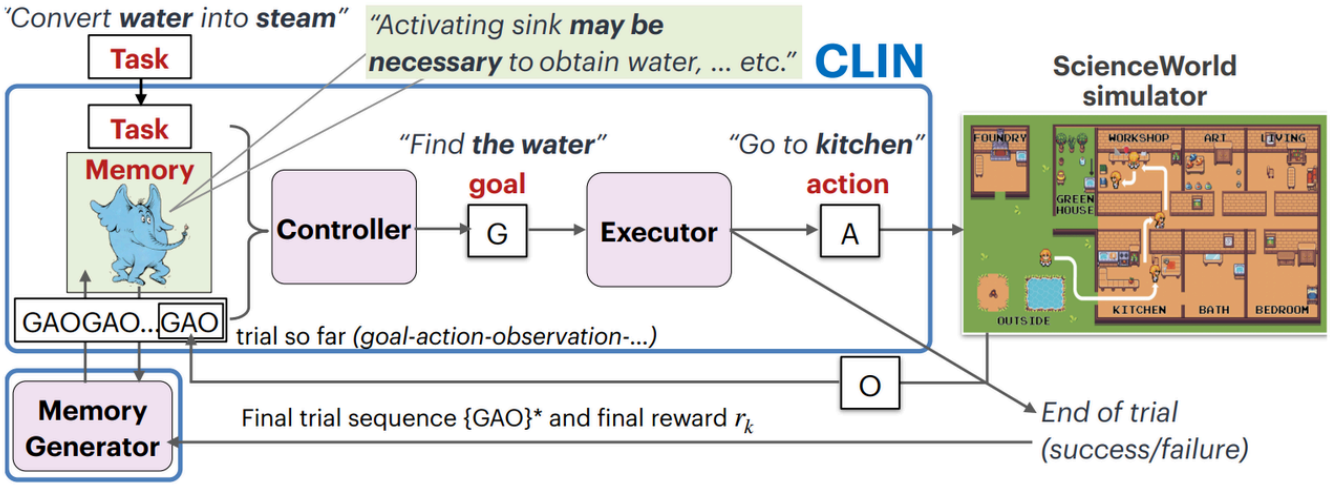
CLIN: A CONTINUALLY LEARNING LANGUAGE AGENT FOR RAPID TASK ADAPTATION AND GENERALIZATION

> #Self-evolve #Agents

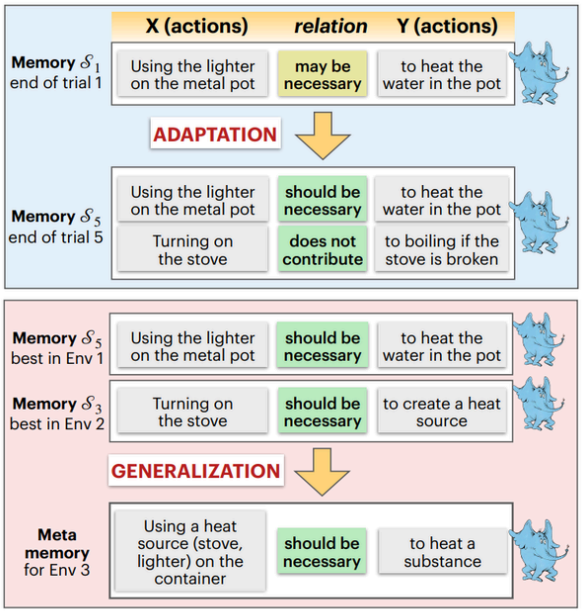


- What problems does this framework solve?
  - (At the time where the paper was written) It is hard for normal LLM agents to learn and self-evolve from mistakes and trials without finetuning.
  - Reflective LLM agents can learn from self-reflection, but their knowledge can only be applied to a specific environment. The generalization ability is poor.
- What does this framework do?

Algorithm 1 Continual Learning with CLIN

procedure ADAPTATION(Task: m, Env: e, Memory: $\mathcal{S}$):
  Initialize Memory: $\mathcal{S}_0$
  for $k \in 1, \cdots, K$ **do**:
    Intialize Trial $\mathcal{T}, t$
    while $t <$ max. steps or task not complete **do**:
      $g_t = \texttt{Controller}\ (m, e, \mathcal{T}_{<t}, \mathcal{S}_{k-1})$
      $a_t = \texttt{Executor}\ (g_t$, admissible actions$)$
      $r_t, o_t = \texttt{Simulator}\ (\mathcal{T}_{<t}, a_t)$
      $\mathcal{T}_{<t+1} = \mathcal{T}_{<t} + (g_t, a_t, o_t, r_t)$
    Final reward $r_k = r_t$
    $\mathcal{S}_k = \texttt{memory-generator}\ (\{\mathcal{S}_{<k}\}, \mathcal{T}_k, r_k)$

procedure GENERALIZATION(Task: m, Env: e, past m'/e')
  $\{\mathcal{S}_{\text{crucial}}, r_k\} = \texttt{crucial-memories}$ (past m'/e')
  $\mathcal{S}_{\text{meta}} = \texttt{meta-memory}\ (\{\mathcal{S}_{\text{crucial}}, r_k\}, m)$
  ADAPTATION(m, e, $\mathcal{S}_{\text{meta}}$)



  - Instead of using finetuning or self-reflection, the paper proposes a RAG-based way to maintain the memory that represents the knowledge the model has learned from interactions with the environment.
  - In the memory, instead of saving experiences, it saves **Causal Abstractions.**

    - The framework contains 2 agents during the interaction with the environment:
        1. Controller: Generate the next sub-goal to achieve the final goal based on the current environment, the task, the trial context, and the memory from the last trail.
        2. Executor: With the sub-goal from the controller and the possible actions of the current state, the executor will generate an action. If the action is not valid, it will find the most similar action using embedding. If the similarity is below a threshold, this record of failure will be added to the context, and the executor will repeat generation until reach the maximum times of try.
        - With the observation from the environment, triplets with `(Sub-goal, action, observation)` will be then added to the trajectory (trial context).
    - Finally, the environment will return a reward to evaluate the quality of the trajectory.
    - A **memory generator** will convert the series of triplets to a memory entry with multiple causal abstractions based on the reward and the memory from the recent 3 trials. It can be a set of: `X is necessary to achieve Y`, `X contributes nothing to Y`.
    - Generalization: To generalize the memory, the framework will select the **best** trajectory for each experience (with multiple trajectories) and use the *best's* to do further abstraction to create **meta-memory** with experience-specific elements removed. Then, whenever there is a new task, the meta-memory will be **adapted** to that environment serve as the a priori.
- Experiment Results
    - The research team conducted tests within the virtual text environment ScienceWorld.
    - Rapid adaptation: Through repeated attempts in the same environment and task, CLIN consistently improved its performance, ultimately achieving scores significantly surpassing the current state-of-the-art reflective agent Reflexion (by 23 absolute points).
    - Robust generalization: CLIN successfully transferred past experience to entirely new environments (zero-shot performance improved by 4 points) and even to entirely new tasks (zero-shot performance improved by 13 points).
    - Ablation experiments demonstrate that downgrading structured “causal abstractions” to unformatted free-text suggestions reduces the agent's average score by 6 points, validating the effectiveness of its memory architecture.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM